MILITARY SPECIFICATION

AMPLIFIERS, INTERMEDIATE FREQUENCY AM–1063/MS AND AM–1064/MS

1. SCOPE

1.1 **Scope.** This specification covers two types of intermediate frequency (IF) amplifier which are part of the moving target indicator (MTI) synchronizer (see 6.2).

1.2 **Classification.** The amplifiers shall be one of the following types:

Type I —Delay amplifier, designated as Intermediate Frequency Amplifier AM–1063/MS, detects and amplifies delayed modulated 15-megacycle (mc) radio frequency (RF) carrier.

Type II —Nondelay amplifier, designated as Intermediate Frequency Amplifier AM-1064/MS, detects and amplifies nondelayed modulated 15-mc RF carrier.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–F–13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.

MIL–P–14232 —Parts Equipment and Tools for Ordnance Materiel; Packaging of.

MIL–I–45208 —Inspection Requirements, General Specification for.

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–109 —Quality Assurance Terms and Definitions.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

D7616994 —Amplifier, Intermediate Frequency AM–1064/MS.

D7617072 —Amplifier, Intermediate Frequency AM–1063/MS.

PACKAGING DATA SHEETS

MIL–P–14232/P7616994 —Packaging of Amplifier, Intermediate Frequency AM–1064/MS.

MIL–P–14232/P7617072 —Packaging of Amplifier, Intermediate

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Frequency AM–1963/MS.

(Copies of specifications standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. The amplifier shall be manufactured in accordance with Drawings D7617072 and D7616994 for Type I and Type II respectively.

3.2 General specification. The amplifiers shall meet the requirements of Specification MIL–F–13926, as specified below. Should any conflict exist between the general specification and this specification, this specification shall take precedence.

(a) Order of precedence

(b) Dimensions and tolerances

(c) Inorganic protective surface finishes

(d) Part identification and marking

(e) Workmanship

3.3 Environmental service conditions.

3.3.1 *Vibration.* The amplifiers shall withstand a vibration of a simple harmonic motion having an amplitude of 0.0075 (0.015 inch total excursion). The frequency shall be varied uniformly between the limits of 10 to 55 cycles per second (cps) and return to 10 cps. Cyclic time shall be applied for a period of 10 minutes in each of the three mutually perpendicular directions (total of 30 minutes). Subsequent to vibration, the amplifiers shall meet the requirements of 3.4.

3.3.2 *Temperature.*

3.3.2.1 *Operating.* The amplifiers shall meet the requirements of 3.4, when thermally stabilized at ambient temperatures of −40° and +150°F.

3.3.2.2 *Storage.* The amplifiers shall meet the requirements of 3.4 at an ambient temperature of between 60° to 90°F. after having been thermally stabilized at ambient temperatures of −60° and +165°F.

3.4 Performance. The functional requirements of 3.4.1 or 3.4.2 shall be exhibited for the respective typeamplifier with the following conditions:

(a) 15-megacycle (mc) carrier voltages shall be applied at Jack J1.

(b) Jack J2 shall be shorted to ground.

(c) A 2000 ohm ±5%, ½ watt (W) resistor shall represent the output load.

(d) Voltages of + 250 volts, direct-current (Vdc) at 55 milliamperes (ma); + 150 Vdc at 50 ma; and 6.3 volts, alternating-current (Vac), 400 cycles per second (cps), at 1.5 amperes (a) shall be applied at the appropriate pin designations of Plug P1.

(e) The amplifiers shall be allowed a warmup period for 15 minutes.

Note. All symbols referred to are those shown on Drawing E7604169 or Drawing F7604170.

3.4.1 *Type I—(delay amplifier).*

3.4.1.1 *Gain.* The output voltage (Eo) of the amplifier shall be 5.0 Vdc minimum with an input voltage (Ei) not exceeding 13 millivolts (mVac).

3.4.1.2 *Bandwith.* At 3 decibles (db) down from the center frequency, the difference between the high and low frequencies shall be 3 mc minimum.

3.4.1.3 *Center frequency.* The arithmetic mean of the high and low frequencies shall be 15 ± 0.2 mc.

3.4.2 *Type II—(nondelay amplifier).*

3.4.2.1 *Gain.* The Eo of the amplifier shall be 5.0 Vdc minimum with an Ei not exceeding 0.2 Vac.

3.4.2.2 *Bandwith.* At 3 db down from the center frequency, the difference between the high and low frequencies shall be 3 mc minimum.

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3.4.2.3 *Center frequency.* The arithmetic mean of the high and low frequencies shall be 15 ± 0.2 mc.

3.5 Stabilization. The amplifiers shall remain stable and adhere to their respective functional requirements (3.4.1 or 3.4.2) after 1 hour of operation at an ambient temperature of between 60° to 90°F.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL–I–45208 and the component inspection requirements of Specification MIL–F–13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL–STD–109

4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" specified in Standard MIL–STD–105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL–F–13926. Examination and tests related to section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL–STD–105, and the inspection level and sampling plans specified in table I titled "Classification of Defects." Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL–P–11232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

TABLE I. *Classification of defects*

Use Inspection Level II of table I with Sampling Plan table II–A of Standard MIL–STD–105.

	Requirement	Test procedure
Critical: None		
Major: AQL 1.0 Percent Defective		
101. Vibration	3.3.1	4.5.2
102. Insufficient Gain		
Type I	3.4.1.1	4.5.4.1
Type II	3.4.2.1	4.5.5.1
103. Incorrect bandwith		
Type I	3.4.1.2	4.5.4.2
Type II	3.4.2.2	4.5.5.2
104. Incorrect center frequency		
Type I	3.4.1.3	4.5.4.3
Type II	3.4.2.3	4.5.5.3
105. Stabilization breakdown	3.5	4.5.6
Minor: None		

4.2.3 *Disposition of nonconforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" specified in Standard MIL–STD–105.

4.3 Control tests. One amplifier shall be selected at random from each 50 produced, or from each months production, whichever occurs first. Control samples shall have success-

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fully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control test	*Requirement*	*Test procedure*
Fabrication	3.1	4.5.1
General Specification	3.2	Visual, MIL–F–13926
Operating temperature	3.3.2.1	4.5.3
Storage temperature	3.3.2.2	4.5.3

4.3.1 *Control test failure.* Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 **Inspection equipment.** Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL–I–45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Government furnished inspection equipment.* Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment," and "Government Furnished Material" requirements of Specification MIL–I–45208.

4.4.2 *Contractor furnished inspection equipment.*

4.4.2.1 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL–F–13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.2 *Devices used for environmental testing.* All environmental testing devices used to determine conformance of the environmental requirements for vibration and temperature shall conform to the "Test Facilities" requirements outlined in Specification MIL–F–13926.

4.4.2.3 *Test equipment.* In addition to the above mentioned requirements the following test equipment shall be used.

(1) Signal Generator	Continuously variable from 10 to 20 megacycles ±5% unmodulated, to operate into unbalanced 75 ohm load, ±0.5% full scale.
(2) Sensitive R.F. voltmeter	1 millivolt to 3 volts, meter accuracy 3%
(3) Voltmeter, D.C.	20,000 ohms per volt, overall scale accuracy ±2%.
(4) Resistor ½ watt 200 ohms	±5%
(5) Power Supply	±0.2 volt
6.3 volts 400 cycles	1.5 ampere
(6) Power Supply +250 volts	± 2 volt 0.55 ampere
(7) Power Supply +150 volts	± 2 volt 0.50 ampere

4.5 **Test methods and procedures.**

4.5.1 *Fabrication.* The amplifier shall be subjected to a visual, tactile and dimensional inspection for conformance to the applicable drawings and requirements of this specification. Dimensional inspection shall be performed using standard measuring equipment.

4.5.2 *Vibration.* This test shall be conducted

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in accordance with Procedure I of Specification MIL–F–13926 except that the motion specified in 3.3.1 shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Performance measurements of the amplifier are not required during vibration testing. Subsequent to vibration, the amplifier shall be subjected to a visual and tactile examination and shall show no evidence of physical failure. The amplifier shall then be subjected to the tests specified in 4.5.4 and 4.5.5 and shall conform to the requirements specified.

4.5.3 *Storage and operating temperatures.* This test shall be conducted in accordance with Procedure I of Specification MIL–F–13926, at the temperature specified in 3.3.2.1 and 3.3.2.2. While stabilized at each temperature specified in 3.3.2.1, the amplifier shall be energized and continuously operated for 1 hour. At the completion of this operating period the amplifier shall be subjected to the tests specified in 4.5.4 or 4.5.5 whichever is applicable, and shall conform to the requirements specified. Upon completion of the specified exposures and return to standard ambient temperature (60 to 90 degrees Fahrenheit) the amplifier shall be inspected for damage and then subjected to the tests specified in 4.5.4 or 4.5.5 and shall conform to the requirements specified.

4.5.4 *Type I—(delay amplifier).*

4.5.4.1 *Gain.* Connect Signal Generator (Item 1 of 4.4.2.3) to J1 of the amplifier. Short J2 to ground. Maintain 15 megacycles on the signal generator with the input voltage as specified in 3.4.1.1 to J1 of the amplifier. Monitor this input voltage throughout entire test with a sensitive R.F. voltmeter (Item 2 of 4.4.2.3). Energize the amplifier by applying +250 volts (55 ma) to pin 1 of the amplifier, +150 volts (50 ma) to pin 2 and 6.3 volts (400 cycles 1.5 a) to pin 4.

Note. The amplifier shall be allowed a warmup period of 15 minutes before testing. Before conducting subsequent tests the same shall apply. The initial warmup period shall suffice for all tests if amplifier is not de-energized.

The voltage output shall be measured from pin 3 to ground across the 2K resistor (Item 4 of 4.4.2.3) with a D.C. voltmeter (Item 3 of 4.4.2.3) and shall be a minimum of 5.0 volts as specified in 3.4.1.1.

4.5.4.2 *Bandwidth.* Utilize the same set-up specified in 4.5.4.1. Energize the amplifier as specified in 4.5.4.1. Vary the frequency output from the signal generator above and below 15 mc until there is a 3 db drop in the output on each side of the center frequency. The difference between these two frequencies is the 3 db bandwidth and shall be at least 3 mc in accordance with the requirements of 3.4.1.2.

4.5.4.3 *Center frequency.* This test shall be made in conjunction with the test specified in 4.5.4.2. The center frequency of the transmission band as measured between the 3 decibel points shall not exceed the limits specified in 3.4.1.3.

4.5.5 *Type II—(nondelay amplifier).*

4.5.5.1 *Gain.* Connect signal generator (Item I of 4.4.2.3) to J1 of the amplifier. Short J2 to ground. Maintain 15 megacycles on the signal generator with the input voltage as specified in 3.4.2.1. Monitor this input voltage throughout entire test with a sensitive R.F. voltmeter (Item 2 of 4.4.2.3). Energize the amplifier by applying +250 volts (55 ma) to pin 1 of the amplifier, +150 volts (50 ma) to pin 2 and 6.3 volts (400 cycles 1.5 a) to pin 4.

Note. The amplifier shall be allowed to warmup period of 15 minutes before testing. Before conducting subsequent tests the same shall apply. The initial warmup period shall suffice for all tests if amplifier is not de-energized.

The voltage output shall be measured from pin 3 to ground across the 2K resistor (Item 3 of 4.4.2.3) with a D.C. voltmeter (Item 3 of 4.4.2.3) and shall be a minimum of 5.0 volts as specified in 3.4.2.1.

4.5.5.2 *Bandwidth.* Utilize the same setup specified in 4.5.5.1. Energize the amplifier as specified in 4.5.5.1. Vary the frequency output from the signal generator above and below 15 mc until there is a 3 db drop in the

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output on each side of the center and shall be at least 3 mc in accordance with the requirements of 3.4.2.2.

4.5.5.3 *Center frequency.* This test shall be made in conjunction with the test specified in 4.5.5.2. The center frequency of the transmission band as measured between the 3 decibel points shall not exceed the limits specified in 3.4.2.3.

4.5.6 *Stabilization.* This test shall be run concurrent with the tests specified in 4.5.4 and 4.5.5, at the temperature specified in 3.5, and shall meet the requirements specified.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking for the amplifiers shall be in accordance with Specification MIL–P–14232 and Packaging Data Sheets as specified below:

Item	*Packaging Data Sheet*
(a) Amplifier, Intermediate Frequency AM–1063/MS	MIL–P–14232/P7617072
(b) Amplifier, Intermediate Frequency AM–1064/MS	MIL–P–14232/P7616994

6. NOTES

6.1 Intended use. As specified below, the two type amplifiers are intended to:

Type I —Detect and amplify the delayed modulated 15-mc RF carrier from the delay cell amplifier

Type II —Detect and amplify nondelay modulated 15-mc RF carrier from the delay cell amplifier

Both amplifier outputs are sent to the MTI synchronizer for comparison between stationary and moving targets.

6.2 Application. The amplifiers are part of the Synchronizer, Electrical AN–140/MS. The amplifiers, as noted, are used in the following systems:

(a) *Amplifier, Intermediate Frequency AM–1063/MS*
  (1) Guided Missile Fire Control System AN/MSG–3
  (2) Antiaircraft Fire Control Systems M33C and M33D
  (3) Nike Ajax
  (4) Nike Hercules

(b) *Amplifier, Intermediate Frequency AM–1064/MS*
  (1) Guided Missile Fire Control System AN/MSG–3
  (2) Antiaircraft Fire Control Systems M33C and M33D
  (3) Nike Ajax

6.3 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

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